

Lecture 5: Parametric Equations / Equations of Lines

We have seen: 1 line = Intersection of 2 planes

Another way: The trajectory of a moving point: "parametric equation"

What is the position at time t , $q(t)$?

$$Q(t) = (x(t), y(t), z(t))$$

$$Q_0 Q(t) = (x(t)-x_0, y(t)-y_0, z(t)-z_0)$$

$$\begin{cases} x(t)+1 = t \\ y(t)-2 = t \\ z(t)-2 = -3t \end{cases} \Leftrightarrow \begin{cases} x(t) = t-1 \\ y(t) = t+2 \\ z(t) = 2-3t \end{cases}$$

$$Q(t) = Q_0 + t \vec{Q_0 Q_1}$$

Example: Line through Q_0, Q_1

$$Q_0 = (-1, 2, 2) \quad Q_1 = (1, 3, 1)$$

$Q(t)$ = moving point at $t=0$ it's at Q_0 , moves at constant speed

only $x, y, z > 1$



Application: Intersection of a plane!

Consider the plane $x+2y+4z=1$

Where does the line through Q_0, Q_1 intersect the plane?

$$Q_0 = (-1, 2, 2) \rightarrow x+2y+4z = (-1)+2 \cdot 2+4 \cdot 2 = 11 > 1 \text{ Not in the plane}$$

$$Q_1 = (1, 3, 1) \rightarrow 1+2 \cdot 3+4 \cdot 1 = 14 > 1 \text{ Not in the plane}$$

Q_0, Q_1 are on opposite sides of the plane

What about $Q(t)$?

$$x(t)+2y(t)+4z(t) = (-1+t) + 2(t+2) + 4(2-3t) = -1+t+2t+4+8-12t = -9t+11$$

$$Q(t) \text{ is in the plane if } -9t+11=1 \Rightarrow t=1/2$$

Then $Q(1/2) = (0, 5/2, 1/2)$ = intersects the plane at this coordinates

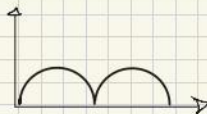
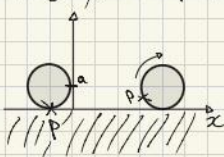
Note: * If line is in the plane then plugging $x(t), y(t), z(t)$ in the equation always gives t .

* If line is parallel to plane, we get another constant

More generally, we can use parametric equations for arbitrary motions in the plane or in space.

Cycloid: Wheel of radius " a " rolling on the floor = x -axis

P = a point on rim of wheel starts at O . What happens?



Question: position $(x(\theta), y(\theta))$ of the point P as a function of the angle θ by which the wheel has rotated

$$\vec{OP} = \vec{OA} + \vec{AB} + \vec{BP}$$

$$\vec{OA} = \langle a\theta, 0 \rangle$$

$$\text{amount by which the wheel has moved } \vec{OA} = a \times \text{length from } A \text{ to } P$$

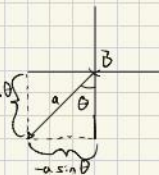
$$\vec{AB} = \langle 0, a \rangle$$

$$\vec{BP} = \langle a \sin \theta, -a \cos \theta \rangle$$

$$\vec{OP} = \langle a\theta - a \sin \theta, a - a \cos \theta \rangle$$

$$x(\theta) = a\theta - a \sin \theta$$

$$y(\theta) = a(1 - \cos \theta)$$



Question: What happens near bottom point?

Take length unit = radius: $a=1$

$$\begin{cases} x(\theta) = \theta - \sin \theta \\ y(\theta) = 1 - \cos \theta \end{cases}$$

$$\begin{cases} \sin(\theta) \sim \theta \\ \text{for a very small } \theta \\ \cos(\theta) \sim 1 \\ \text{for very small } \theta \end{cases}$$

Taylor approximation: for t small, $F(t) \approx F(0) + F'(0)t + \frac{t^2}{2}F''(0) + \frac{t^3}{6}F'''(0) + \dots$

$$\sin \theta \approx \theta - \frac{\theta^3}{6} \quad \cos \theta \approx 1 - \frac{\theta^2}{2}$$

$$x(\theta) \approx \theta - (\theta - \frac{\theta^3}{6}) \approx \frac{\theta^3}{6}$$

$$y(\theta) \approx 1 - (1 - \frac{\theta^2}{2}) \approx \frac{\theta^2}{2}$$

$$\frac{y}{x} \approx \frac{\theta^2/2}{\theta^3/6} = \frac{3}{\theta} \rightarrow \infty \text{ when } \theta \rightarrow 0$$

Slope at origin is ∞ :

